

```
===== test session starts =====
platform darwin -- Python 3.11.15, pytest-9.1.1, pluggy-1.6.0 -- /Users/tuananhnguyen/Documents/Uni/Project Management/hcmus-projectmanage--lab/src/backend/.env/bin/python3
cachedir: .pytest_cache
rootdir: /Users/tuananhnguyen/Documents/Uni/Project Management/hcmus-projectmanage--lab/src/backend
configfile: pyproject.toml
plugins: anyio-4.14.2
collected 142 items

tests/api/test_auth.py::test_dev_token_issues_valid_jwt_with_role PASSED [ 0%]
tests/api/test_auth.py::test_dev_token_defaults_to_reader_role PASSED [ 1%]
tests/api/test_auth.py::test_dev_token_rejects_unknown_role PASSED [ 2%]
tests/api/test_auth.py::test_dev_token_disabled_when_mock_auth_off PASSED [ 2%]
tests/api/test_auth.py::test_register_creates_reader_and_returns_jwt PASSED [ 3%]
tests/api/test_auth.py::test_register_rejects_duplicate_email PASSED [ 4%]
tests/api/test_auth.py::test_register_rejects_disallowed_domain PASSED [ 4%]
tests/api/test_auth.py::test_register_accepts_any_domain_by_default PASSED [ 5%]
tests/api/test_auth.py::test_register_rejects_short_password PASSED [ 6%]
tests/api/test_auth.py::test_login_returns_jwt_for_correct_credentials PASSED [ 7%]
tests/api/test_auth.py::test_login_rejects_wrong_password PASSED [ 7%]
tests/api/test_auth.py::test_logout_returns_no_content PASSED [ 8%]
tests/api/test_auth.py::test_google_login_returns_404_when_not_configured PASSED [ 9%]
tests/api/test_auth.py::test_google_login_redirects_when_configured PASSED [ 9%]
tests/api/test_auth.py::test_google_callback_issues_jwt_with_reader_role_for_new_user PASSED [ 10%]
tests/api/test_auth.py::test_google_callback_reuses_existing_users_role PASSED [ 11%]
tests/api/test_categories.py::test_create_parent_child_and_read_two_level_tree PASSED [ 11%]
tests/api/test_categories.py::test_category_name_parent_and_depth_validation PASSED [ 12%]
tests/api/test_categories.py::test_rename_and_delete_category PASSED [ 13%]
tests/api/test_categories.py::test_delete_rejects_categories_with_children_or_documents PASSED [ 14%]
tests/api/test_categories.py::test_assigning_unknown_category_returns_clear_404 PASSED [ 14%]
tests/api/test_categories.py::test_category_writes_rejected_for_reader_and_editor_roles PASSED [ 15%]
tests/api/test_categories.py::test_category_read_stays_open_without_a_token PASSED [ 16%]
tests/api/test_documents.py::test_upload_creates_document_and_pending_ocr_job PASSED [ 16%]
tests/api/test_documents.py::test_upload_rejected_without_token PASSED [ 17%]
tests/api/test_documents.py::test_upload_rejected_for_reader_role PASSED [ 18%]
tests/api/test_documents.py::test_invalid_upload_returns_422_without_side_effects[malware.exe-MZ-application/octet-stream] PASSED [ 19%]
tests/api/test_documents.py::test_invalid_upload_returns_422_without_side_effects[wrong.pdf-%PDF-1.4-image/png] PASSED [ 19%]
tests/api/test_documents.py::test_invalid_upload_returns_422_without_side_effects[fake.png-not a png-image/png] PASSED [ 20%]
tests/api/test_documents.py::test_invalid_upload_returns_422_without_side_effects[fake.jpg-not a jpeg-image/jpeg] PASSED [ 21%]
tests/api/test_documents.py::test_source_read_back_and_not_found PASSED [ 21%]
tests/api/test_documents.py::test_private_document_hidden_from_guest PASSED [ 22%]
tests/api/test_documents.py::test_public_document_readable_by_guest PASSED [ 23%]
tests/api/test_editor.py::test_get_and_update_page_text_persists PASSED [ 23%]
tests/api/test_editor.py::test_page_update_allows_empty_text PASSED [ 24%]
tests/api/test_editor.py::test_page_update_rejected_for_reader_role PASSED [ 25%]
tests/api/test_editor.py::test_page_update_rejected_for_editor_who_does_not_own_document PASSED [ 26%]
tests/api/test_editor.py::test_page_update_allowed_for_editor_who_owns_document PASSED [ 26%]
tests/api/test_editor.py::test_missing_document_or_page_returns_clear_404 PASSED [ 27%]
tests/api/test_health.py::test_health_returns_ok PASSED [ 28%]
tests/api/test_health.py::test_metrics_returns_prometheus_data PASSED [ 28%]
tests/api/test_highlights.py::test_create_then_list_returns_highlight PASSED [ 29%]
tests/api/test_highlights.py::test_list_is_empty_before_any_highlight PASSED [ 30%]
tests/api/test_highlights.py::test_overlapping_highlights_both_persist PASSED [ 30%]
tests/api/test_highlights.py::test_identical_range_creates_separate_record PASSED [ 31%]
tests/api/test_highlights.py::test_deleting_document_removes_highlights PASSED [ 32%]
tests/api/test_highlights.py::test_other_user_cannot_see_highlights PASSED [ 33%]
tests/api/test_highlights.py::test_anonymous_cannot_create_highlight PASSED [ 33%]
tests/api/test_highlights.py::test_blank_selection_is_rejected PASSED [ 34%]
tests/api/test_highlights.py::test_cross_chapter_range_is_rejected PASSED [ 35%]
tests/api/test_highlights.py::test_cannot_create_highlight_on_draft_document PASSED [ 35%]
tests/api/test_highlights.py::test_existing_highlights_readable_after_unpublish PASSED [ 36%]
tests/api/test_highlights.py::test_create_on_missing_document_returns_404 PASSED [ 37%]
tests/api/test_highlights.py::test_note_survives_reload PASSED [ 38%]
tests/api/test_highlights.py::test_note_can_be_edited PASSED [ 38%]
tests/api/test_highlights.py::test_clearing_note_keeps_the_highlight PASSED [ 39%]
tests/api/test_highlights.py::test_note_longer_than_limit_is_rejected PASSED [ 40%]
tests/api/test_highlights.py::test_other_user_cannot_edit_note PASSED [ 40%]
tests/api/test_highlights.py::test_overlapping_highlights_keep_separate_notes PASSED [ 41%]
tests/api/test_highlights.py::test_delete_removes_highlight_permanently PASSED [ 42%]
tests/api/test_highlights.py::test_delete_leaves_overlapping_highlight_intact PASSED [ 42%]
tests/api/test_highlights.py::test_other_user_cannot_delete_highlight PASSED [ 43%]
tests/api/test_highlights.py::test_acting_on_already_deleted_highlight_returns_404 PASSED [ 44%]
tests/api/test_highlights.py::test_delete_works_after_unpublish PASSED [ 45%]
tests/api/test_metadata.py::test_metadata_is_trimmed_saved_and_returned_by_document_detail PASSED [ 45%]
tests/api/test_metadata.py::test_optional_metadata_can_be_null_or_blank PASSED [ 46%]
tests/api/test_metadata.py::test_required_metadata_reports_each_blank_field PASSED [ 47%]
tests/api/test_metadata.py::test_metadata_document_not_found PASSED [ 47%]
tests/api/test_metadata.py::test_metadata_rejected_for_reader_role PASSED [ 48%]
tests/api/test_metadata.py::test_metadata_rejected_for_editor_who_does_not_own_document PASSED [ 49%]
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